

Risk Factors For the Indian Equity Market: Statistics, Visualization, and an Interactive Tool

Aman Foujdar, Sandeep Juneja, Atish Kumar, and Nagpurnanand Prabhala.*

June 13, 2026

Abstract

We develop a carefully specified set of six Fama–French style risk factors for the Indian equity market in the 21st century. We also provide tools for researchers and practitioners to examine portfolio characteristics and compositions, as well as interactive visualizations to assess the dynamics of factor premiums, cumulative returns across several horizons, and to compare a flexible set of long-short strategies. We provide a transparent documentation of the protocols and the filters used to generate the factor premiums. The factor library is publicly available at factorboosting.github.io

JEL Codes: G12, G11, G14, G17

Keywords: : Factor investing, asset pricing, Fama-French Factors, Emerging markets .

*Aman Foujdar and Sandeep Juneja are at Ashoka University and can be reached at aman.foujdar_ug2023@ashoka.edu.in and sandeep.juneja@ashoka.edu.in, respectively. Atish Kumar is at the Indraprastha Institute of Information Technology Delhi and can be reached at atishk@iiitd.ac.in. Juneja is Professor of Computer Science and Director, Safexpress Centre for Data, Learning, and Decision Sciences at Ashoka University. Nagpurnanand Prabhala is at Carey Business School, The Johns Hopkins University. Email: prabhala@jhu.edu. We thank the CMIE for providing the financial data and several helpful discussions.

1 Introduction

The Fama–French factor framework has become a central tool in empirical asset pricing. It offers a parsimonious set of factors that represent sources of systematic risk. The early work (Fama and French, 1993) supplemented the classic market factor in CAPM with two additional factors, viz., size, and value, to which Carhart (1997) introduced a momentum factor. In later work, Fama and French (2016) introduced the profitability and investment factors. The six risk factors are foundational for both academic research and portfolio management in practice, although other factors – and alternative measures of the factors - have emerged and continue to emerge.¹

Factor models are widely used in academic research and practice. The models serve as asset pricing benchmarks for new anomalies to test whether they genuinely generate alpha. The benchmarks help attribute performance of mutual fund and pension fund managers or advisors providing portfolio management services. Factor premiums also serve as targets for mutual funds and ETFs that offer factor investing products to their clientele. Understanding the factor risk premiums is also of economic interest. For example, finance research attempts to understand whether momentum in stock returns reflects factor momentum (Ehsani and Linnainmaa, 2022). In corporate finance, they serve as benchmarks to test post-announcement portfolio and calendar time returns. Factor models also improve cost of capital estimates used by managers to evaluate projects.

Our goal is to develop reliable factor estimate for the six core Fama-French factors for the Indian equity market. India is a large emerging economy. Like many non-U.S. markets, India has distinct signatures in market development, participation, regulations, and trading. A stock market was established in 1875 as The Native Share & Stock Brokers’ Association, now called the BSE. Operations continued past Indian independence under strict control through the Controller of Capital Issues. A major shift occurred in 1992-1994 on the heels of stock market scandals with the establishment of the National

¹For example, Pastor and Stambaugh (2003) consider a liquidity factor, for which Liu (2006) and Amihud (2002) propose alternative metrics. The range of factors has led to what has been labeled as the “factor zoo.” See, e.g., Harvey, Liu and Zhu (2016) and Feng, Giglio and Xiu (2020).

Stock Exchange. A system of electronic screen-based trading was introduced at the time, a transition to which was essentially completed around 2000. Our estimates of factor premiums begin around this time.

Despite its popularity, there is limited academic literature in the Indian context. The most notable work has been by Agarwalla, Jacob and Varma (2014) (JRV henceforth) where they constructed Fama-French 3 (FF3) and momentum factors for the stocks listed on the Bombay stock exchange. They have recently updated the universe of stocks to include both NSE and BSE stocks along with some additional filters and are updated thrice every year. Despite applying filters, they consider many small and not frequently traded firms as a part of their universe. A more recent study by Jensen, Kelly and Pedersen (2023) (JKP henceforth) constructed 153 factors across 13 themes for 93 countries. In order to remain consistent across countries, they apply a uniform methodology to produce these anomalies. A major drawback of this approach is that it undermines the market structure of each country and ignores the investability aspect of the stocks. Since a vast majority of the stocks listed on the Indian stock exchanges are small and illiquid and hence the portfolio returns may not be representative of the market. In short, the existing sources suffer from either one or more of the following shortcomings, first, inclusion of small and illiquid stocks for factor construction, secondly, limited coverage of factors, third ignoring the nuances of the Indian equity markets and lastly they are not updated monthly which can be used by investors.

In this paper, we propose a rigorous and empirically grounded approach to solve for these problems, our methodology ensures that the idiosyncrasies of the Indian equity markets are captured. We have extended the factor coverage by adding Investment, Operating Profitability. To ensure that the factor returns are reflective of the tradable universe in the Indian markets, we have performed careful analysis of the market structure before applying filters. The factor returns are provided at three levels: entire tradable universe, top 300 and top 500 stocks by market capitalization.

2 Data for Factor Construction

The firm-level balance sheet, daily stock market variables and share suspension data are sourced from the CMIE Prowess DX database. We extracted the daily stock market variable along with the annual balance sheets of publicly traded companies listed on the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE). The time period for our analysis spans over 25 years, starting from April 2000 and updated monthly. There are a total of 8113 unique firms in our entire dataset.

2.1 Portfolio Formation

The entire factor construction has been organized under portfolio years. In our construction, a Portfolio Year (PY) starts from October of current year to September of next year. Concretely, for a trading day t , we define the PY in the following manner.

$$PY(t) = \begin{cases} \text{year}(t) + 1, & \text{if } \text{month}(t) \in \{10, 11, 12\}, \\ \text{year}(t), & \text{if } \text{month}(t) \in \{1, \dots, 9\}. \end{cases}$$

The reason behind this convention is that a vast majority of the Indian companies have their financial end year in March and the financial results are often announced with a few months lag. Following the Fama and French (1993) methodology, we take a six month lag and form portfolios on end September. In contrast, Jensen, Kelly and Pedersen (2023) takes a four month lag instead of six. The breakpoint calculation for each factor and the holding universe for the next portfolio year is constructed using the data from the previous portfolio year.

2.2 Exchange selection

The Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE) are India's leading stock exchanges. The BSE has the largest number of listed companies among all

exchange in India. The NSE on the other hand was founded in 1992 to introduce modern trading practices and leads in terms volume and total market capitalization. Companies are often listed on both the exchanges and this creates a challenge of exchange selection for factor construction. In our base data, there are 4150 and 633 firms which exclusively listed on the BSE and NSE respectively. Although the number of firms listed exclusively on NSE are comparatively lower than BSE, but there has been a sharp rise in recent years. There are 2974 companies which are listed on both the exchanges. **Figure 1** shows firms listings on NSE and BSE over the years, even though BSE dominates in terms of firm listings but exchange rate listing percentage share is declining and NSE's share is rising. The exchange decision is made annually based on aggregate traded value in the preceding Oct–Sep year (PY t), then applied to all daily rows in the holding year (PY $t + 1$). The decision is shifted forward by one year to avoid look-ahead bias. We follow a simple rule of exchange selection, if a stock has been listed on the NSE then we take the NSE values else we choose BSE. This approach is line with the market trends as NSE has emerged as the dominant player in terms of volume.

3 Filters for Indian Equity Markets

Any reliable factor construction methodology must pay attention to the composition of the investable universe. Inclusion of stocks which are practically harder to own and trade may distort factor premia. These concerns are particularly severe in the Indian equity market, where a significant chunk of listed stocks are practically non-tradable. Infrequent trading and low market capitalization do not allow investors to seamlessly take any meaningful positions on such stocks. To counter this, our methodology is formulated in manner explicitly addresses these issues. In this section, we discuss how the filters have been applied to on the raw data to arrive at the tradable universe.

3.1 Negative Book Equity

Book equity is defined as the total shareholders' equity reported in a firm's balance sheet. It is calculated by taking the difference of total assets and total liabilities. Firms may encounter a scenario where their total liabilities exceeds the total assets, even though the firms are not necessarily insolvent, negative book equity affects the interpretability of factors. For example a negative book-to-market ratio has no meaningful economic interpretation in the context of Value factor. We exclude all firms with negative book equity from our holding universe. Specifically, we compute book equity using the most recently available balance sheet data as of March 31 of the portfolio formation year, applying a six-month reporting lag to avoid look-ahead bias. A firm is excluded from the universe for portfolio year $t + 1$ if its book equity as of March 31 of year t is either missing or strictly negative. **Figure 2** shows the yearly count of firms with negative book equity in the raw data. An important point to note here is that, we drop firms with negative book equity only for those factors which don't require book equity variable for construction.

3.2 Penny Stock Filter

3.2.1 Overview

We define a particular stock a penny stock if its median daily closing share price, in the preceding one year of the portfolio formation date is less than or equal to Rs. 10. We remove all the penny stocks from our holding universe in the next portfolio year in order to ensure the stocks used for factor construction are investable and don't exhibit volatile with relatively small volumes of trades.

3.2.2 Why do we need a penny stock filter?

Pandey and Sehgal (2016) argue that penny stocks are highly speculative and have limited liquidity which can lead to price manipulations. Additionally, such stocks show extreme

volatility in price with large swings in either directions. These characteristics make such stocks both expensive to trade, as well as undesirable to use when tracking performance of macro variables like factors over time, since they could skew the sample.

Moreover, penny stocks have vastly lower oversight from retail and institutional investors, meaning that data collated about them is relatively less verified, and thus may contain a significant number of mistakes or aberrations.

3.2.3 What is a good choice for the filter and why?

The U.S Securities and Exchange Commission (SEC) has an official definition of penny stocks, stock below \$5 share price are classified as penny stocks. However, the Securities and Exchange Board of India (SEBI) has not issued a similar definition for penny stocks in India or any minimum trading price which could have been used as a proxy. Several studies focusing on the Indian equity markets have defined penny stocks Pandey and Sehgal (2016) uses a cutoff of Rs. 10. However, Agarwalla, Jacob and Varma (2014) use a penny stock threshold of Re. 1. Jegadeesh and Titman (2023) in their cross country study involving India has used a cutoff of \$1.

Figure 3 & Figure 4 show that the percentage of firms with share price less than Rs. 10 remained relatively stable from 2005 - 2020 at 25-30% of the total firms. On the other hand, the percentage of firms with share price less than Rs. 30 dropped uniformly after 2020, probably due to inflows into the equity markets and the resultant broad-based growth in market caps. The percentage of stocks with share price below Re. 1 is extremely small across sample years, using Re. 1 as the cutoff will not significantly alleviate the three above mentioned problems which arise when considering penny stocks.

Table 1 shows also that the 25th percentile of the annual share price distribution hovered around Rs. 10 from 2005 till 2020, post which it climbed to around Rs. 27.

Thus, given the stability of the Rs. 10 cutoff over the vast majority of the sample period and a cutoff of Re. 1 captures very few firms, we believe that Rs. 10 is an appropriate ceiling to use in order to define a penny stock.

3.3 Microcap Filter

In addition to the penny stock filter, we also apply the microcap filter. The goal of this filter is to drop stocks with low market capitalization which are often harder to trade despite them not being a penny stock. We follow the Agarwalla, Jacob and Varma (2014) approach to define the microcap cutoff which involves dropping all the firms whose market capitalization on the last traded day of a particular stock in that PY is less than the 10 percent of the median market capitalization. **Figure 5** shows the microcap cutoff for each PY and in the past five years has remained between Rs. 50-70 Crores.

3.4 Liquidity Filter

There is a sizable presence of illiquid firms, inclusion of such stocks in the portfolio may distort estimated premia. Discontinuities in trading can stem from various multiple factors such as suspensions, lack of supply or demand for the shares. In order to ensure that the factor returns are not influenced by illiquid firms we drop illiquid firms from the factor calculation pipeline. This treatment ensure that the returns are representative of the tradable universe only.

We call a firm illiquid on the portfolio formation day if the firm has not traded every full trading week atleast once in the preceding PY. A trading week is defined as the week starting from Monday and ending on Friday. If a firm is listed during the year then the liquidity calculation is applied only from the week it started trading. There exists other approaches of applying the liquidity filter. Notably, Agarwalla, Jacob and Varma (2014) consider a firm illiquid if it has traded less than 50 days in the preceding year. One of the problems with their criteria is that there can be cases where firms may pass the liquidity threshold while exhibiting prolonged periods of illiquidity. Our criteria is more stringent and considers only those companies which are highly liquid in nature.

3.5 Firms in holding universe by PY

After applying the filters mentioned above, we arrive at the tradable universe for our approach. An important point to note here is that for arriving at size breakpoints, we don't impose the positive book equity restriction. Due to this, the universe for Size breakpoint is slightly larger than the universe for other factors. Since, a significant chunk of the firms with negative book equity overlap with other filters. **Table 2** shows the holding universe by each PY. **Table 3** reports the number of additional firms excluded due to negative book equity in each portfolio year.

4 Factor Construction

The factors constructed in this paper follow the double-sort methodology standard in asset pricing literature with adjustments made for the Indian setting. Each stock is assigned factor label as per the definition and the stocks are sorted into size and specific factor groups to form portfolios. These portfolios are then used to compute long-short factor returns. In this section, we elaborate on the methodology to define long-short returns for each factor.

4.1 Factor Formulas

4.1.1 Size: Annually rebalanced

The size factor is constructing using the market capitalization of firms. Following standard practice, we use the September end-of-month market capitalization to assign size labels for the subsequent portfolio year. Firms above the 90th percentile of market capitalization are classified as Big (B), the remaining firms are classified as Small (S). This is a departure from the original Fama-French methodology where the median market capitalization was considered to split Big and Small firms. To account for the microstructure of the Indian equity markets which are characterised by many small firms and most of

the market capitalization is concentrated in the top 10 percentile. This is in line with the existing work in the Indian context.

4.1.2 Book-to-Market (B/M): Annually rebalanced

The book-to-market ratio is a valuation signal which compares firm's market equity with its market value. Book equity is measured as of March 31 of the formation year, applying a six-month reporting lag to ensure there is no look-ahead bias. For firms announcing their financial results after March 31, they are considered for next year's factor construction.

$$B/M_t = \frac{\text{Book Equity}_t}{\text{Market Cap}_t} \quad (1)$$

Firms with a B/M ratio above the 70th percentile are classified as Value (V); firms below the 30th percentile are classified as Growth (G), the remaining are Neutral (N). Firms with negative or missing book equity are excluded from value breakpoint construction and holding universe.

4.1.3 Operating Profitability: Annually rebalanced

Operating profitability measures how efficiently a firm generates profit from its day-to-day operations, divided by its lagged book equity. The numerator subtracts cost of goods sold (COGS), selling and distribution expenses, and interest expense from total sales (S&D expense). Lagged book equity (year $t - 1$) is used as the denominator to avoid look-ahead bias.

$$\text{OpProf}_t = \frac{\text{Sales}_t - \text{COGS}_t - \text{S\&D Expenses}_t - \text{Interest Expense}_t}{\text{Book Equity}_{t-1}} \quad (2)$$

Firms above the 70th percentile are classified as Robust (R), firms below the 30th percentile are classified as Weak (W) and the rest are Neutral (N).

4.1.4 Investment: Annually rebalanced

The investment factor captures asset growth of a firm, it is defined as the change in total assets over the two most recent financial years excluding the current year available at the time of portfolio formation in line with the Fama-French style construction. High asset growth firms are classified as Aggressive (A) which considered as a negative signal and while low growth firms are classified as Conservative (C) which is a positive signal.

$$\text{Investment}_t = \frac{\text{Total Assets}_{t-1} - \text{Total Assets}_{t-2}}{\text{Total Assets}_{t-2}} \quad (3)$$

Firms above the 70th percentile and below 30th percentile are classified as Aggressive (A) and Conservative (C) respectively. The remaining firms are Neutral (N).

4.1.5 Momentum: Monthly rebalanced

We define momentum as the cumulative gross return over an 11-month formation window excluding the most recent month. For each firm in a given month t , monthly returns are first computed by compounding daily returns within each calendar month.

The momentum signal is then defined as the cumulative gross return over months $t - 12$ through $t - 2$, skipping month $t - 1$:

$$\text{MOM}_t = \prod_{i=2}^{12} (1 + r_{t-i}) - 1 \quad (4)$$

where the product runs over 11 monthly returns. A firm must have valid returns for 11 months for the signal to be computed, months with missing daily return data are treated as zero-return months within the monthly compounding step. Breakpoints are computed monthly at the 30th and 70th percentiles of the cross-sectional distribution of MOM_t . Firms above the 70th percentile are assigned to the label Winner (W) and firms below the 30th percentile are classified as Loser (L). The remaining firms are assigned the label Neutral(N).

Since momentum is updated monthly, we separately calculate the monthly Size breakpoints using the end of month market capitalization. The Size split methodology remains the same as annual one.

4.1.6 Market Factor

The market factor return is defined as the excess return on the broad market portfolio over the risk-free rate. We use the monthly return on the NIFTY 500 index as a proxy for the market portfolio, given its comprehensive coverage of large, mid and small cap stocks. The risk-free rate is essentially the 91-day Treasury bill rate published on the Reserve Bank of India website.

4.2 Portfolio Construction Methodology

Portfolios are rebalanced annually at the end of September each year. For each portfolio year t , defined as October of year t to September of year $t + 1$, the eligible universe is determined as follows.

4.2.1 Breakpoint Calculation

Breakpoints are calculated annually (for fundamental factors) and monthly (for momentum/size) at three levels, entire tradable universe, Top 300 and Top 500. **Table 4** shows the breakpoints of various factors

4.2.2 Rebalancing Schedule

- **Fundamental Factors (Value, Inv, OpProf):** Signals calculated on financials from March 31, Year T . Portfolios formed on **October 1, Year T** and held until **September 30, Year $T + 1$** .
- **Momentum / Size:** Rebalanced **Monthly**.

4.3 Return Calculation Logic

Portfolios are intersected by **Size** and the **Factor** of interest (e.g., Small-Value, Big-Winner). Depending on the factor we construct either four or six portfolio each (2 Size $\times$ 2/3 Factor Buckets).

4.3.1 Weighting Schemes

- **Value-Weighted (VW):**

$$w_{i,t} = \frac{\text{Market Cap}_{i,t-1}}{\sum \text{Market Cap}_{j,t-1}} \quad (5)$$

- **Equal-Weighted (EW):**

$$w_{i,t} = \frac{1}{N_t} \quad (6)$$

4.4 Long-Short Factor Construction

4.4.1 Portfolio Intersections

For each factor, the eligible universe is classified into six portfolios by leveraging the two Size labels (Small and Big). Size labels are assigned annually using the September end-of-month market capitalization breakpoint, while characteristic labels are assigned using the corresponding annual or monthly breakpoints described in the previous section. This yields six value-weighted portfolios for each factor:

$$\{S, B\} \times \{F_L, F_N, F_H\} \quad (7)$$

where S and B denote Small and Big by market capitalization, and F_L , F_N , F_H denote the low, middle, and high for each factor. Portfolio returns are equal/value-weighted using each firm's previous month's market capitalization as weights. We leverage the constructed portfolio to come up with long-short factor returns, these returns are

represent market neutral premium for each of the factors.

4.4.2 Size Factor (SMB)

The Size factor (Small Minus Big) is computed by taking the average returns of three small portfolios with three big portfolios computed within the value universe:

$$\text{SMB} = \frac{1}{3}[(SV - BV) + (SN - BN) + (SG - BG)] \quad (8)$$

where SV , SN , SG denote the Small-Value, Small-Neutral, and Small-Growth portfolios respectively, and BV , BN , BG denote their Big counterparts. Historically, the Big Value portfolio had thin volumes often less than 5 firms, this trend has been changed in the past decade. Any portfolio with few firms may exhibit idiosyncratic returns, to circumvent this issue, we apply a rule where we don't consider any portfolio which has less than 5 firms for long-short factor construction. This is a standard practice in the literature as well. This restriction amounts to a modification in the factor return premium calculation. For example, if BV had less than 5 firms the BV and SV portfolios are dropped and SMB is computed using only the neutral and growth pairs.

$$\text{SMB} = \frac{1}{2}[(SN - BN) + (SG - BG)] \quad \text{if } n_{BV} < 5 \quad (9)$$

4.4.3 Value Factor (HML)

The Value factor (High-Minus-Low book-to-market) is the average return of the two value portfolios minus the average return of the two growth portfolios:

$$\text{HML} = \frac{1}{2}[(SV - SG) + (BV - BG)] \quad (10)$$

The long portfolios are Small-Value and Big-Value and the short ones are Small-Growth and Big-Growth. When $n_{BV} < 5$, HML is essentially the difference between

small value and small growth:

$$\text{HML} = SV - SG \quad \text{if } n_{BV} < 5 \quad (11)$$

4.4.4 Momentum Factor (WML)

The momentum factor (Winner-Minus-Loser) is the average return of the two winner portfolios minus the average return of the two loser portfolios:

$$\text{WML} = \frac{1}{2} [(SW - SL) + (BW - BL)] \quad (12)$$

where SW and BW denote the Small-Winner and Big-Winner portfolios and SL and BL denote the Small-Loser and Big-Loser portfolios. Momentum and size breakpoints are computed monthly for the momentum factor.

4.4.5 Operating Profitability Factor (RMW)

The operating profitability (Robust-Minus-Weak) factor along with Investment are rely only on the balance sheet variables. It is the difference between average return of the two robust portfolios and the average return of the two weak portfolios:

$$\text{RMW} = \frac{1}{2} [(SR - SW) + (BR - BW)] \quad (13)$$

where SR and BR denote the Small-Robust and Big-Robust portfolios and SW and BW denote the Small-Weak and Big-Weak portfolios. Operating profitability labels are assigned annually using March financial year-end data with a six-month reporting lag.

4.4.6 Investment Factor (CMA)

The investment factor (Conservative-Minus-Aggressive) is the average return of the two conservative portfolios minus the average return of the two aggressive portfolios:

$$\text{CMA} = \frac{1}{2}[(SC - SA) + (BC - BA)] \quad (14)$$

where SC and BC denote the Small-Conservative and Big-Conservative portfolios and SA and BA denote the Small-Aggressive and Big-Aggressive portfolios. Conservative firms are those with low asset growth (below the 30th percentile) and aggressive firms are those with high asset growth (above the 70th percentile).

Table 5 summarises the long and short factor construction.

5 Returns

In this section, we compare our factor performance against the factor series provided by JKP and AJV. Table 6 reports the factor return statistics for the period starting from October 2003 to September 2024 for all the three methodologies discussed in the paper. The performance of the size factor in the 21st century has been a story of two halves. In the first half till 2014, the size factor significantly underperformed and delivered close to 3% CAGR return, however it witnessed a sharp turnaround post 2014 and posted a return of 5.35% CAGR. The Size factor constructed by AJV has followed a similar trend but substantially lower in level and delivers only 0.86% of CAGR return. The value factor premia has delivered a factor premia of 3.92% as per our methodology, on the other hand AJV value premium stood at 10.16%. The performance of the value factor between the two methodologies has been comparable as the difference in the firm universe has narrowed. In the last five years (2021-2026), the value factor for both the methodologies has grown over 250%. The JKP value premium stood at a meagre 0.6% CAGR largely driven by their factor construction methodology.

The momentum factor has been known to perform well in the Indian equity markets and this is evident in all the three cases. Our construction of momentum shows a 10.32% CAGR with a sharpe ratio 0.58, the AJV construction delivers close to 3 percentage

points higher returns. The difference is higher in the initial year and narrows significantly post 2014 and drops below 1 percentage point. The JKP momentum stands at 2.37% with a sharpe ratio of 0.24.

The investment and operating profitability factors have provided stable returns with lower volatility as compared to other factors as per our construction. The investment factor (CMA) provided 5.1% annualized premia with the highest sharpe ratio of 0.61 as per our construction. The Operating Profitability factor (RMW) exhibited stable returns across sub-periods with 4% CAGR over the entire period which is substantially higher than JKP with 1.15% CAGR.

6 Comparison with existing Factor libraries

As a part of constructing the factor library, we drew inspiration and extensively compared our methodology against the existing resources. As mentioned above, Agarwalla, Jacob and Varma (2014) and Jensen, Kelly and Pedersen (2023) publish factors for the Indian equity markets. Agarwalla, Jacob and Varma (2014) focuses solely on the Indian equity markets and their methodology aims to capture the market microstructure but the factor coverage remains limited.

6.1 Comparison against Agarwalla, Jacob and Varma (2014)

Our universe construction departs from Agarwalla, Jacob and Varma (2014) on three dimensions. First, on exchange selection, since it is common for stocks to be listed on multiple exchanges, it is important to have a rule for selecting the exchange from where the prices are sourced. We follow a simple rule, choose NSE if the stock is listed both the exchanges and BSE otherwise. This essentially means that the variables are selected for PY $t + 1$ by applying this rule in PY t . This rule is aimed to reflect the prominence of NSE as an exchange. On the other hand, JRV selects the exchange with higher aggregate

liquidity, though the exact liquidity variable is not specified in their website, we replicated their results with traded value, traded quantity and found that results are largely stable across the two variables. Second, for the liquidity filter, our methodology requires a firm to trade at least once in every full calendar week of the preceding portfolio year. JRV applies a 50-day trading threshold in the preceding portfolio year with no additional restrictions. This may lead to inclusion of firms which have lumpy trading activity, making it less ideal for investment.

Third, our penny stock threshold is Rs. 10 based on median price over the preceding portfolio year as opposed to Re. 1 for Agarwalla, Jacob and Varma (2014). Fourth, the microcap filter which is applied on firms below 10% of the median market capitalization on the portfolio formation date and lastly, the book equity treatment are identical across both methodologies which excludes firms with negative or missing book equity.

6.1.1 Firm Counts and Coverage

Figure 6 shows the number of firms in our universe, the Agarwalla, Jacob and Varma (2014) universe, and the full CMIE Prowess base (firms with a valid identifier) for each portfolio year from 2001 to 2025. Our filters are substantially stricter in the initial years. In 2001, we retained 858 firms against Agarwalla, Jacob and Varma (2014) 2,027 where the total base was 3,539. The gap narrows post-2010 as the market grew but our there is still a considerable difference between the two universes.

6.1.2 Impact of Exclusions

An obvious question to ask is what is the impact of impact of stricter filters as compared to Agarwalla, Jacob and Varma (2014). This can be understood in a variety of ways such as what is the difference in market capitalization and trading volume. **Figure 7** shows the difference in the total September end market capitalization of the Agarwalla, Jacob and Varma (2014) universe, our universe, and the excluded firms for each portfolio year. Excluded firms account for 0.6–8.0% of total market capitalization, with the peak

occurring in 2007. In the post-2013 period the excluded percentage stays below 2.1%. A similar story emerges for trading volume as well, excluded firms account for under 10% of annual trading volume at their peak and typically 1–3% post-2013 (**Figure 8**). These figures support the claim that excluded firms are economically peripheral and that their prices are likely to be unreliable for return inference.

6.1.3 Characteristics of Excluded Firms

Figure 9 reports median market capitalisation for our universe, Agarwalla, Jacob and Varma (2014), and excluded firms by portfolio year. The excluded firms’ median market cap is approximately 10% of our universe throughout the sample period precisely the threshold set by the microcap filter confirming that the filter operates as intended. The median market cap of excluded firms remains below Rs. 30 crore for most of the sample, rising only modestly in recent years as overall market levels increased.

6.2 Comparison against Jensen, Kelly and Pedersen (2023)

There is a structural difference in factor construction methodology when we compare our methodology against Jensen, Kelly and Pedersen (2023). Their Global Factor Data are constructed using CRSP, Compustat, and Compustat Global identifiers and characteristics. For each factor construction, they form monthly portfolios by dividing stocks into terciles using breakpoints computed from only non-micro stocks for each country. They define non-micro cap stocks as firms with market equity above the NYSE 20th percentile. Microcap stocks are then assigned factor label using factor breakpoints computed from the non-micro stocks.

Portfolio returns are computed using three methods, equal-weighted, value-weighted and capped value weights. For the capped value-weighted method they cap the highest market equity to be at the NYSE 80th percentile, and factor returns are defined as high-minus-low long-short portfolios. On the other hand, our factor library sources data from CMIE Prowess with Indian-market-specific sample filters. In the absence of a common

identifier between CMIE Prowess and the WRDS database, we matched firms using a multi-stage fuzzy name-matching procedure, the first level included cleaned full names after removing generic suffixes such as *Ltd.* and *Industries*. In the second level we used two and three token name prefixes, and finally on spaceless normalised names to handle spelling variants and after this procedure. After doing this, there were 113 unique firms remain unmatched.

6.2.1 Impact of exclusions

Figure 10 compares firm counts between our universe and Jensen, Kelly and Pedersen (2023) by portfolio year. Post-2012, Jensen, Kelly and Pedersen (2023) contains close to twice as many firms as our universe. We analysed underlying reasons for such a large deviation in the universe construction and found that the exclusions can be explained by the filters. Since 2013, over 95% of the firms which were a part of the Jensen, Kelly and Pedersen (2023) universe-only fail at least one of the filters. This shows that the differences in the tradable universe can be attributed to differences in methodology. Their factor library doesn't apply filters to account of the nuances of the Indian markets. **Figure 11** shows the unexplained residual of 50-66 firms per year is stable and is due to the 113 firms that could not be matched across CMIE and WRDS identifiers.

The characteristics of Jensen, Kelly and Pedersen (2023) - only firms further validate our exclusions. As shown in **Figure 12**, the median price of firms belonging only to Jensen, Kelly and Pedersen (2023) is ₹12 versus ₹87 for our universe, with the distribution of their firms centred precisely at the ₹10 penny threshold. **Figure 13** shows that JKP-only firms have a median of 178 trading days per year versus 248 for our universe which is significantly lower. **Figure 14** reports median market capitalisation for our universe and JKP-only firms, the latter's median stays substantially below our universe.

The evidence supports our firm exclusions criteria since it removes firms that are small and illiquid which are harder to invest in and hence their inclusion in return measurement may lead to unreliable estimates. These firms represent under 2% of total market

capitalization and under 3% of total trading volume in the post-2013 period.

7 Conclusion

We have constructed a comprehensive factor library for Indian equity markets spanning close to 25 years. Apart from covering the popular Fama-French and Momentum factors, we have constructed additional factors which could be of interest for practitioners and academicians. We have conducted extensive analysis to compare against the existing factor libraries. Our factor library draws inspiration from the existing resources and attempts to make a methodological improvement in the Indian context. Additionally, we aim to leverage these factors to answer other interesting questions.

References

- Agarwalla, Sobhesh Kumar, Joshy Jacob, and Jayanth Rama Varma. 2014. “Four factor model in Indian equities market.” *Indian Institute of Management, Ahmedabad Working Paper*, , (2013-09): 05.
- Amihud, Yakov. 2002. “Illiquidity and Stock Returns: Cross-Section and Time-Series Effects.” *Journal of Financial Markets*, 5(1): 31–56.
- Carhart, Mark M. 1997. “On persistence in mutual fund performance.” *The Journal of finance*, 52(1): 57–82.
- Ehsani, Sina, and Juhani Linnainmaa. 2022. “Factor Momentum and the Momentum Factor.” *Journal of Finance*, 77(3): 1877–1919.
- Fama, Eugene F, and Kenneth R French. 1993. “Common risk factors in the returns on stocks and bonds.” *Journal of financial economics*, 33(1): 3–56.
- Fama, Eugene F, and Kenneth R French. 2016. “Dissecting anomalies with a five-factor model.” *The Review of Financial Studies*, 29(1): 69–103.
- Feng, Guanhao, Stefano Giglio, and Dacheng Xiu. 2020. “Taming the Factor Zoo: A Test of New Factors.” *Journal of Finance*, 75(3): 1327–1370.
- Harvey, Campbell, Yan Liu, and Heqing Zhu. 2016. “... and the Cross-section of R_xpected Returns.” *Review of Financial Studies*, 29: 5–68.
- Jegadeesh, Narasimhan, and Sheridan Titman. 2023. “Momentum: Evidence and insights 30 years later.” *Pacific-Basin Finance Journal*, 82: 102202.
- Jensen, Theis Ingerslev, Bryan Kelly, and Lasse Heje Pedersen. 2023. “Is There a Replication Crisis in Finance?” *The Journal of Finance*, 78(5): 2465–2518.

- Liu, Weimin. 2006. “A Liquidity-Augmented Capital Asset Pricing Model.” *Journal of Financial Economics*, 82(3): 631–671.
- Pandey, Asheesh, and Sanjay Sehgal. 2016. “Explaining size effect for Indian stock market.” *Asia-Pacific Financial Markets*, 23(1): 45–68.
- Pastor, Lubos, and Robert Stambaugh. 2003. “Liquidity Risk and Expected Stock Returns.” *Journal of Political Economy*, 111(3): 642–683.

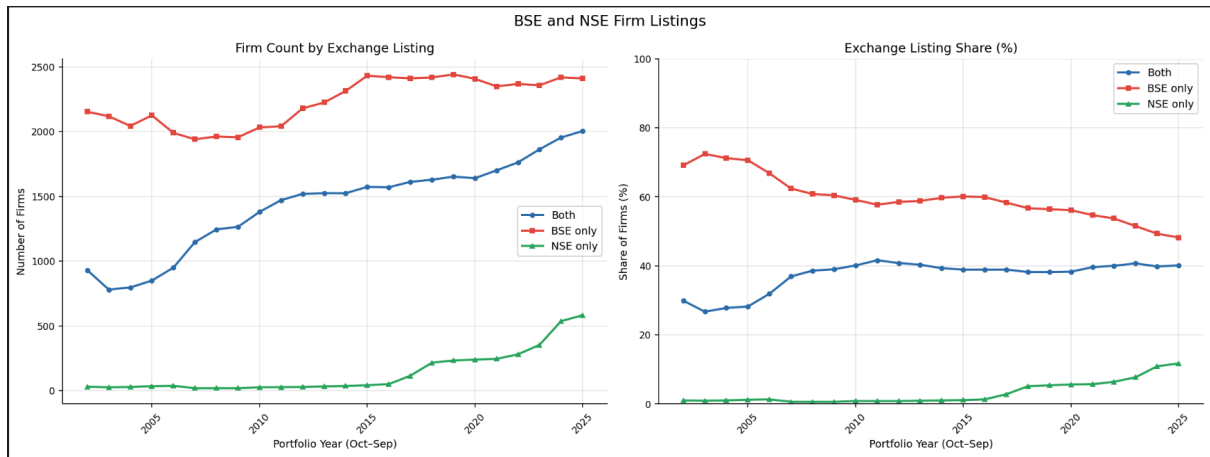


Figure 1: NSE and BSE listings over the years

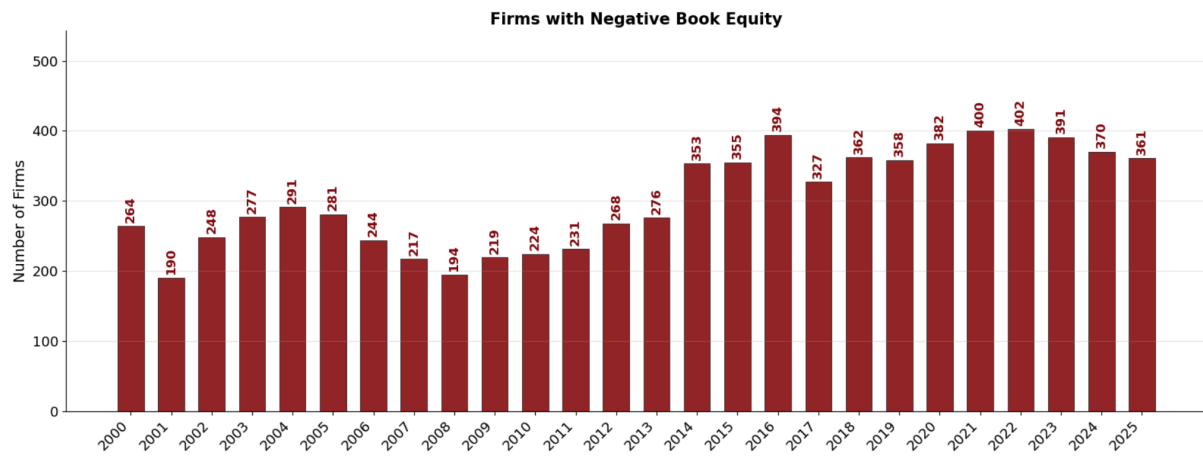


Figure 2: Firms with negative Book Equity as of Portfolio Formation date

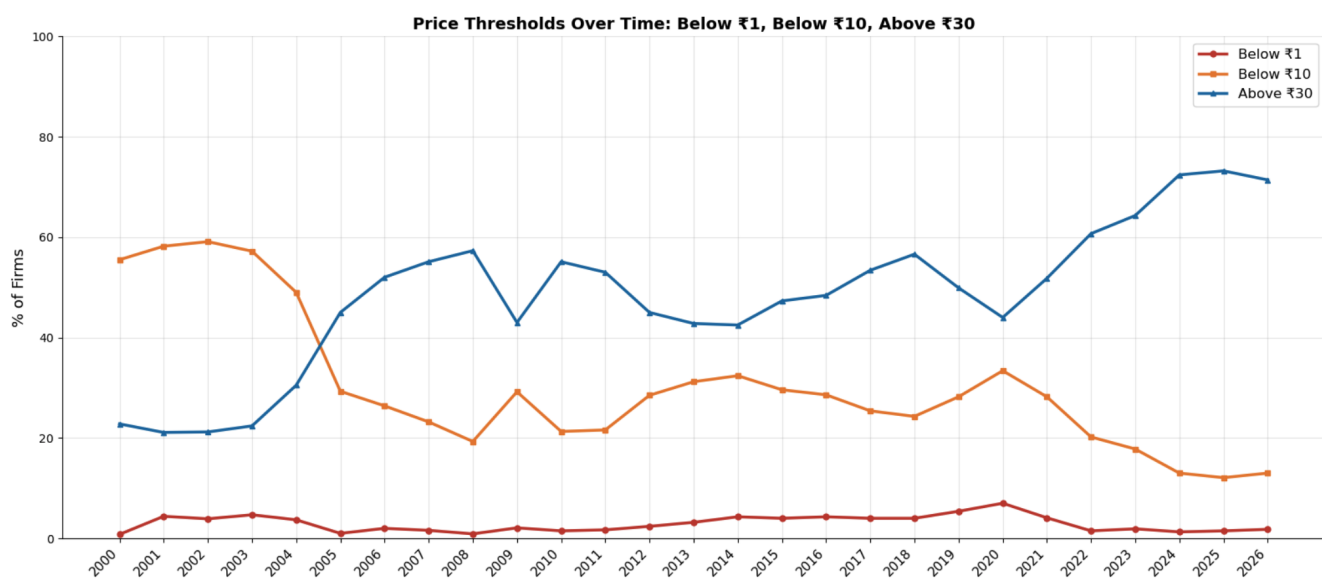


Figure 3: Percentage of firms in each price bucket, PY2000-PY2025

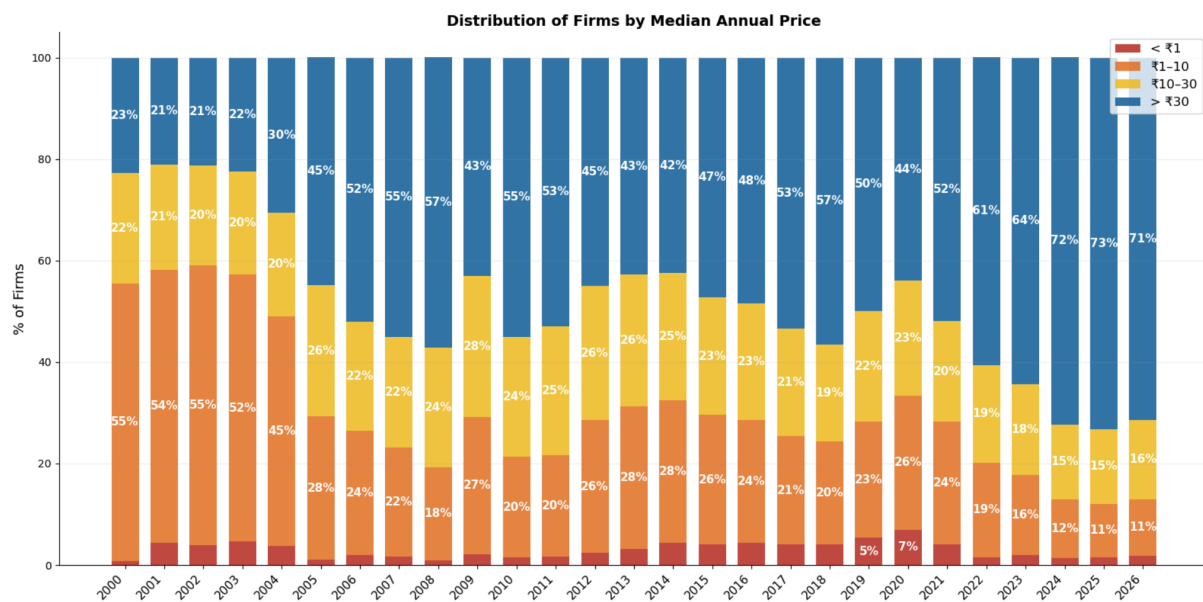


Figure 4: Distribution of firms in each price bucket, PY2000-PY2025

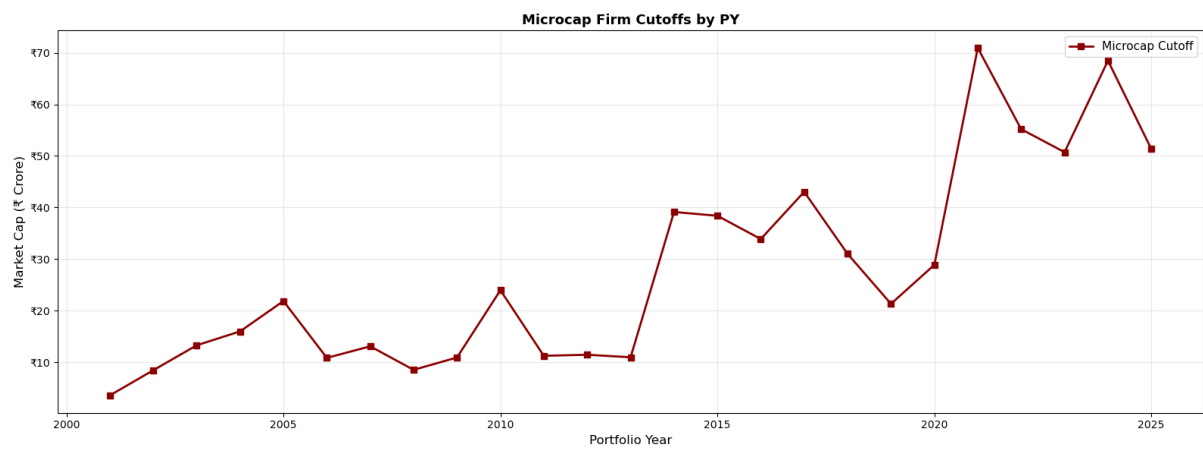


Figure 5: Microcap Cutoff in Crores

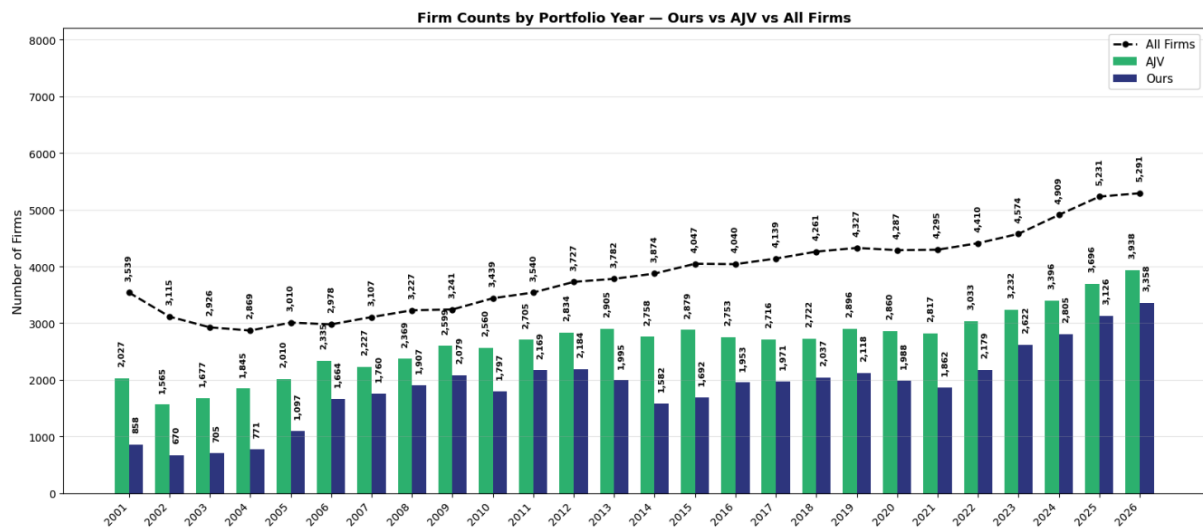


Figure 6: Firm counts by portfolio year. Black dotted line represents the total CMIE prowess base, the maroon and blue bars represent Our and AJV (Agarwalla, Jacob and Varma (2014)) tradable universe each portfolio year.

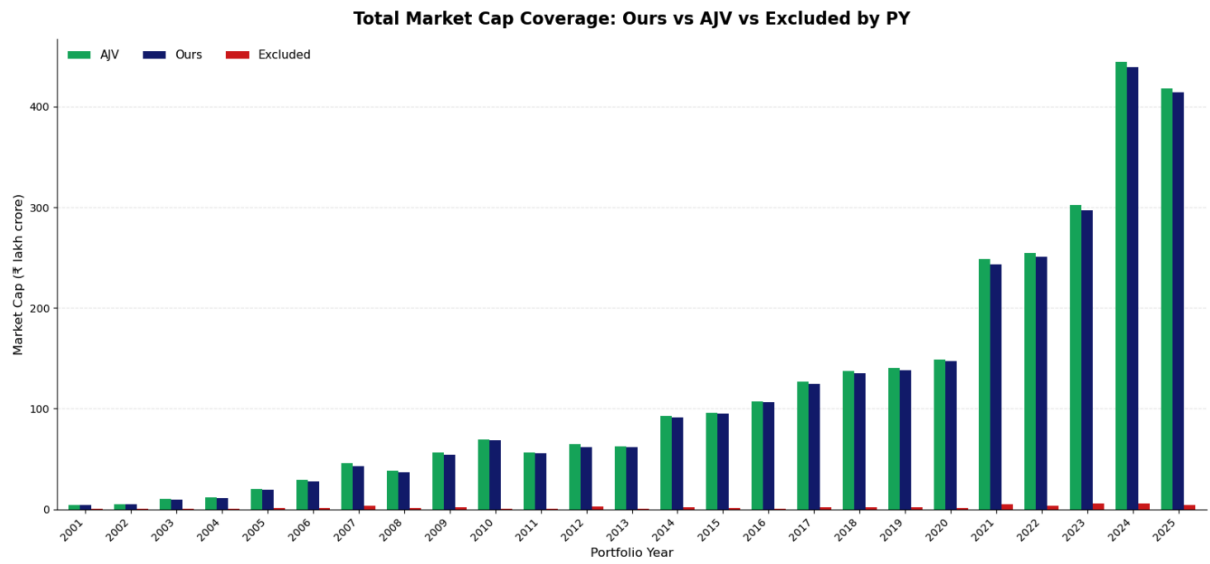


Figure 7: PY denotes portfolio year, defined as October of year t to September of year $t + 1$.

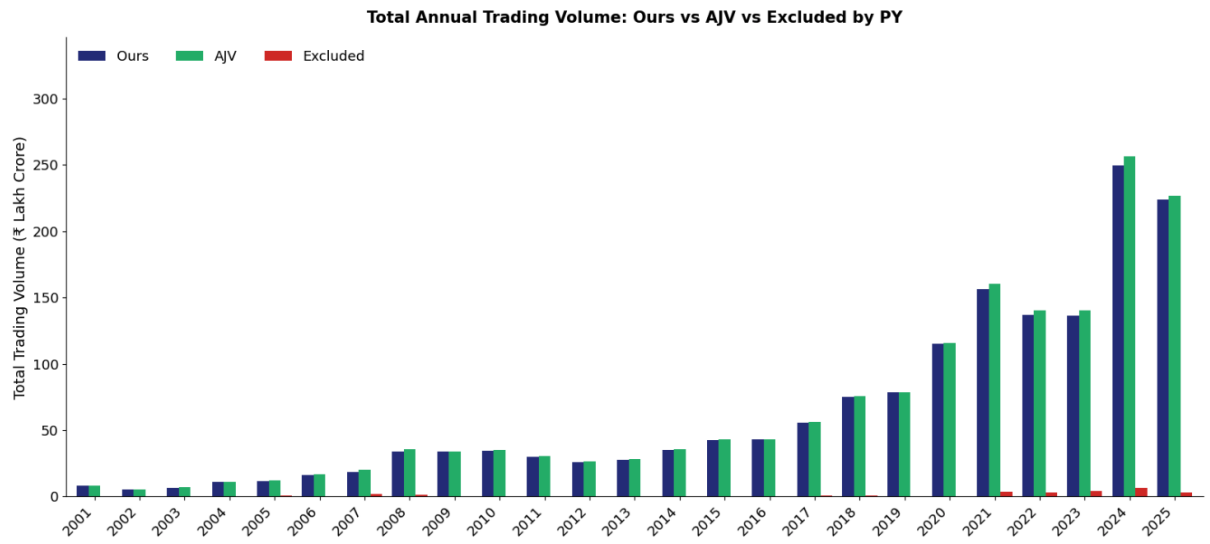


Figure 8: PY denotes portfolio year, defined as October of year t to September of year $t + 1$.

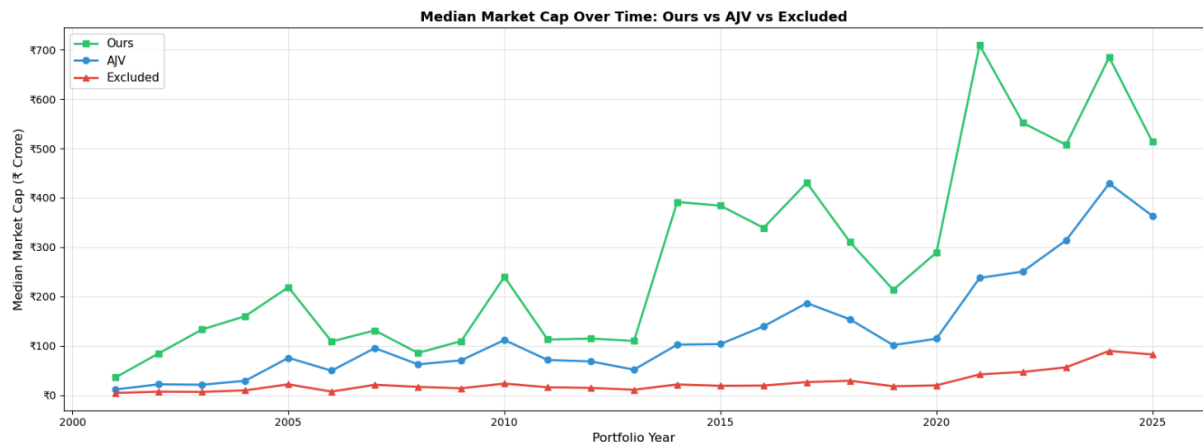


Figure 9: PY denotes portfolio year, defined as October of year t to September of year $t + 1$.

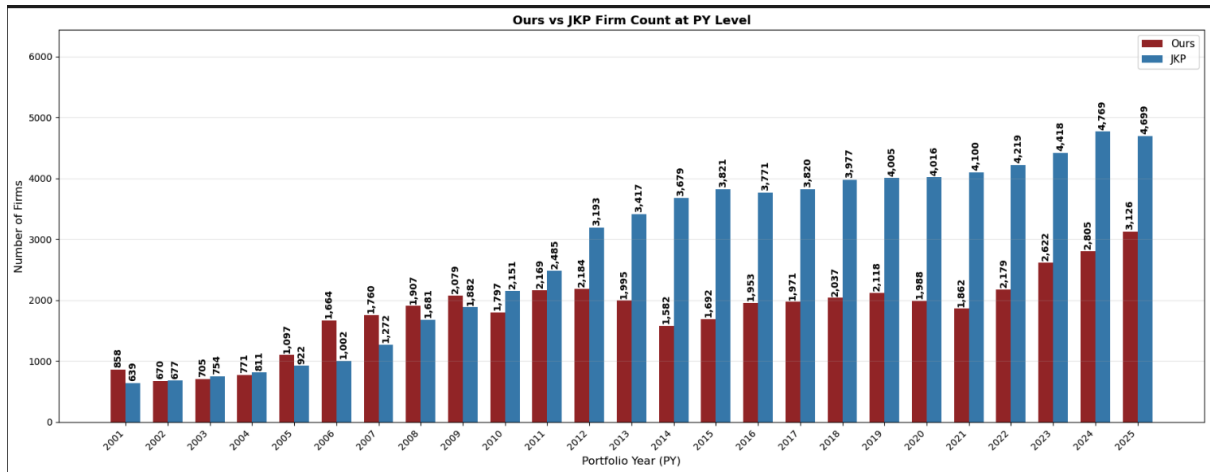


Figure 10: The red and blue bars represent the number of firms in the holding universe using Our and JKP’s methodology respectively

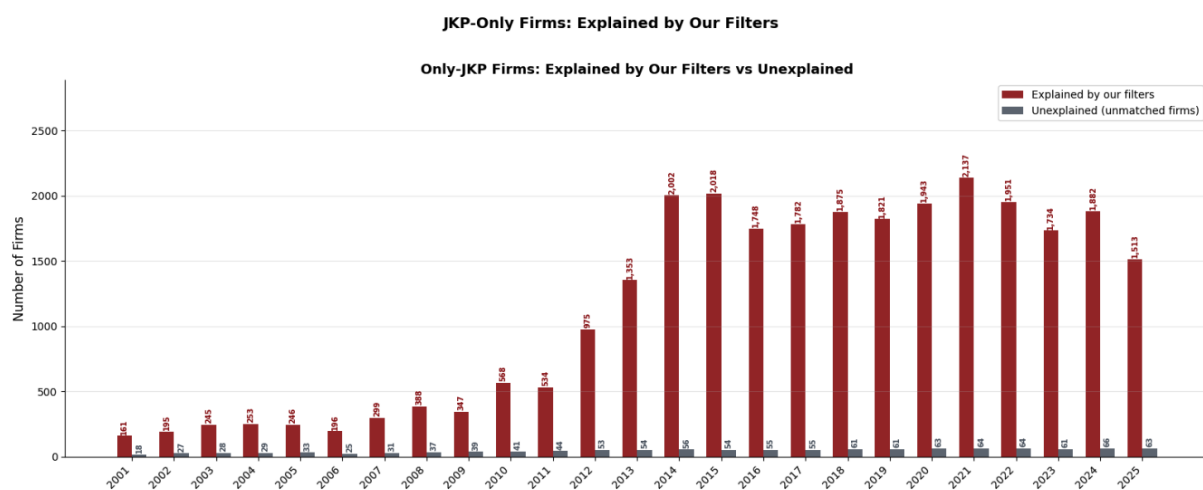


Figure 11: The red bars represent the numbers of firms are filtered from our exclusions criteria but included in JKP universe, the black bars are the unmatched/unexplained firms

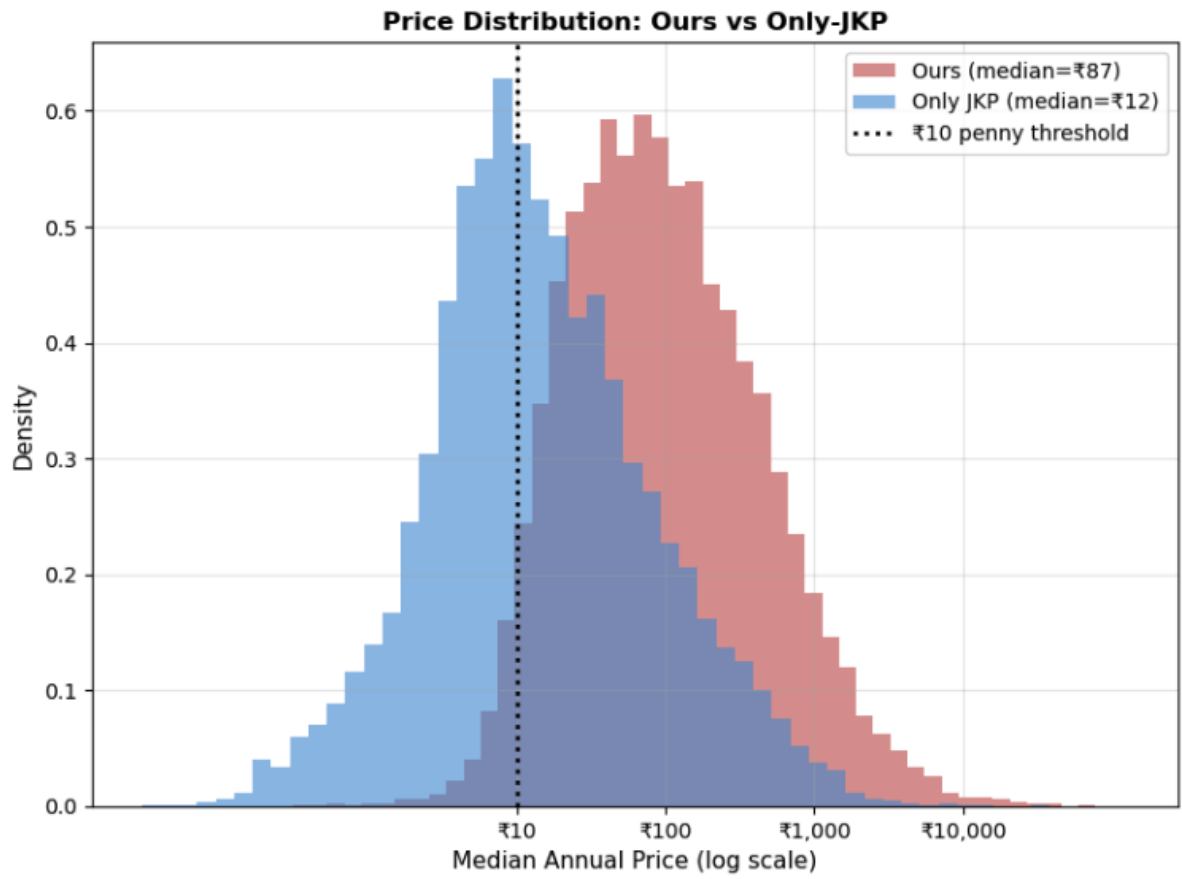


Figure 12: The red and blue bars show the distribution of median prices for Our and JKP universe

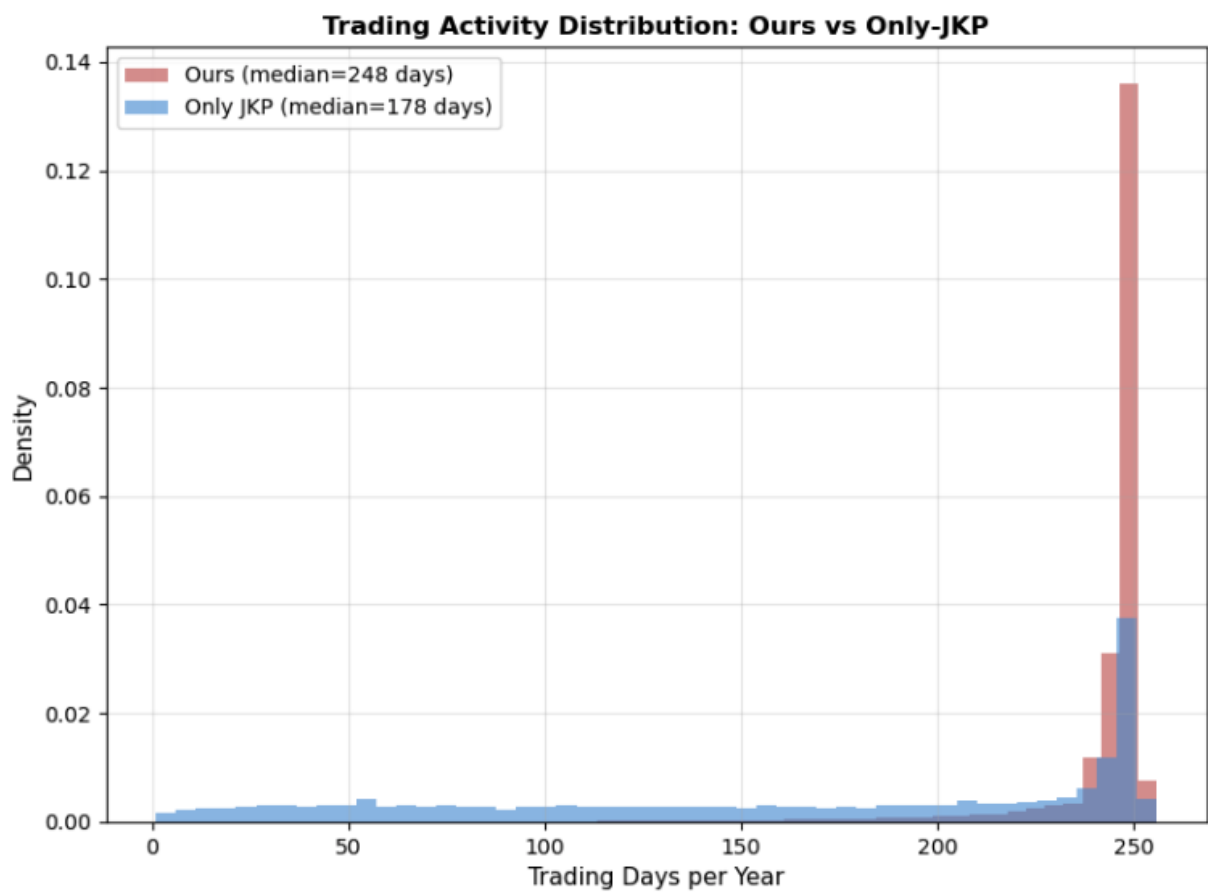


Figure 13: The red and blue bars show the distribution of trading days for Our and JKP universe

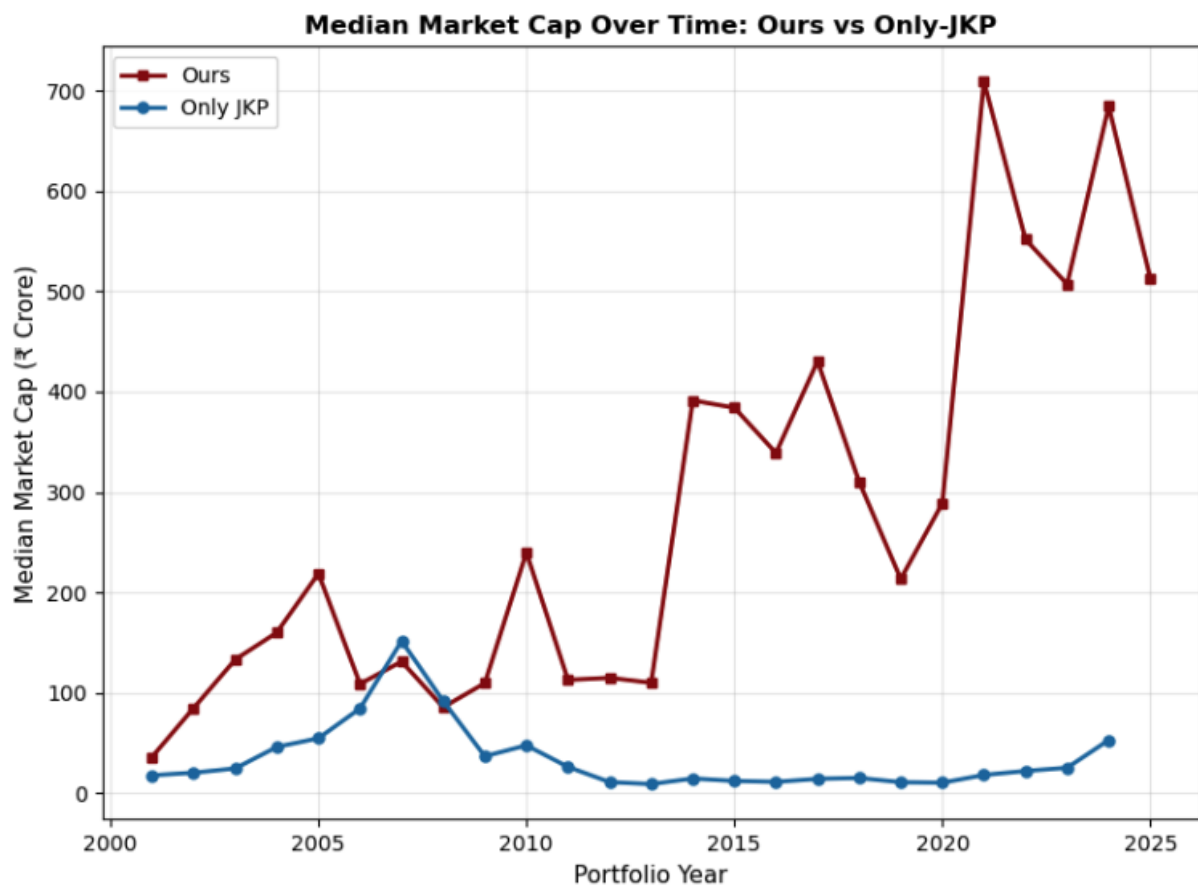


Figure 14: The red and blue lines represent the median market capitalization of firms part of Our and JKP's holding universe

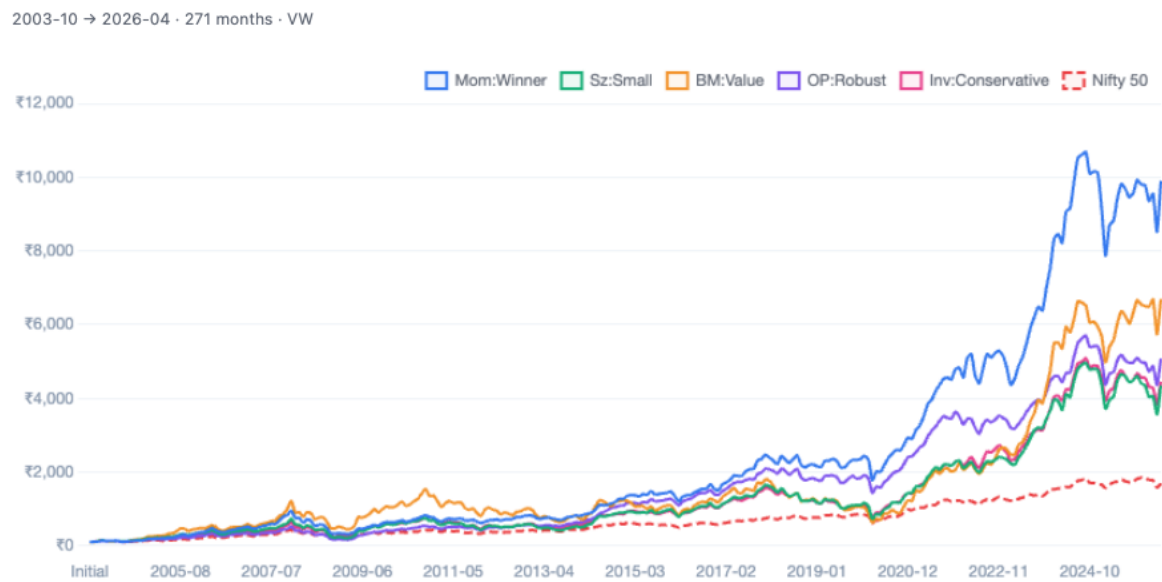


Figure 15: Each line represents long only cumulative wealth starting from Rs.100 using value weighted methodology

2003-10 → 2026-04 · 271 months · VW

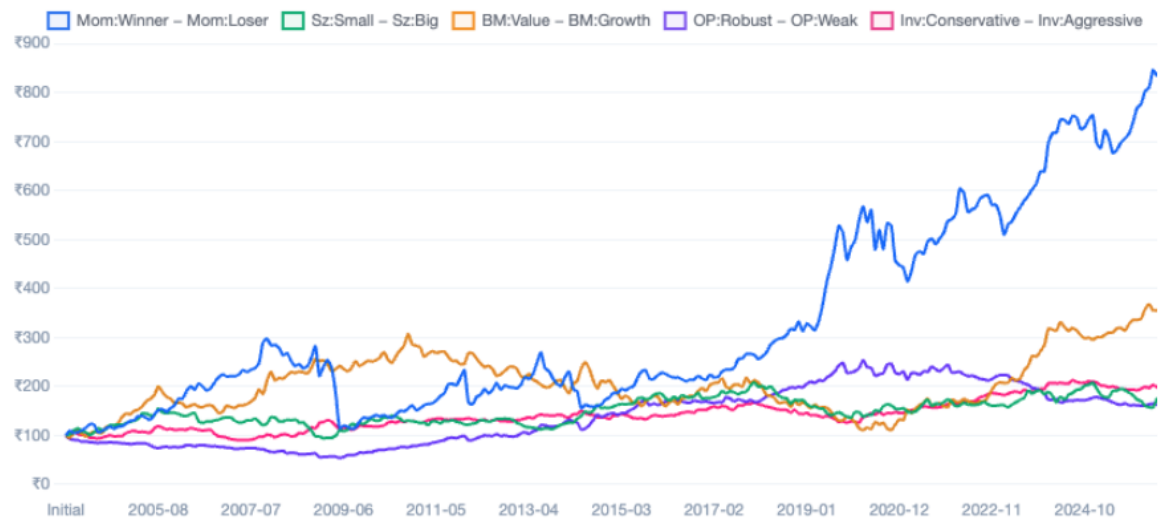


Figure 16: Each line represents long-short cumulative wealth starting from Rs.100 using value weighted methodology

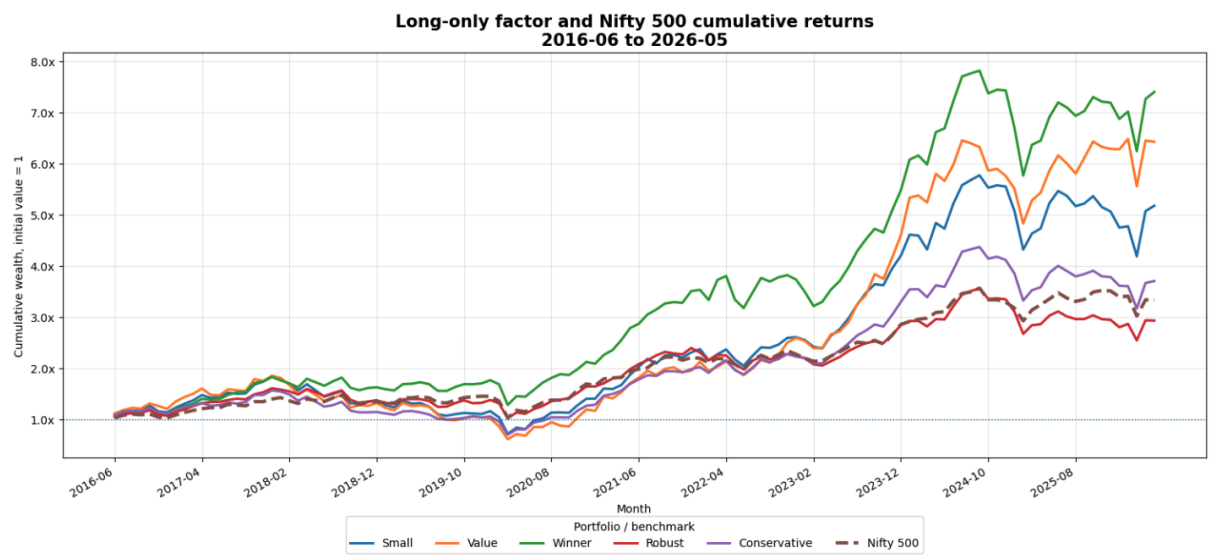


Figure 17: Each line represents long only cumulative wealth starting from ₹1 using value weighted methodology

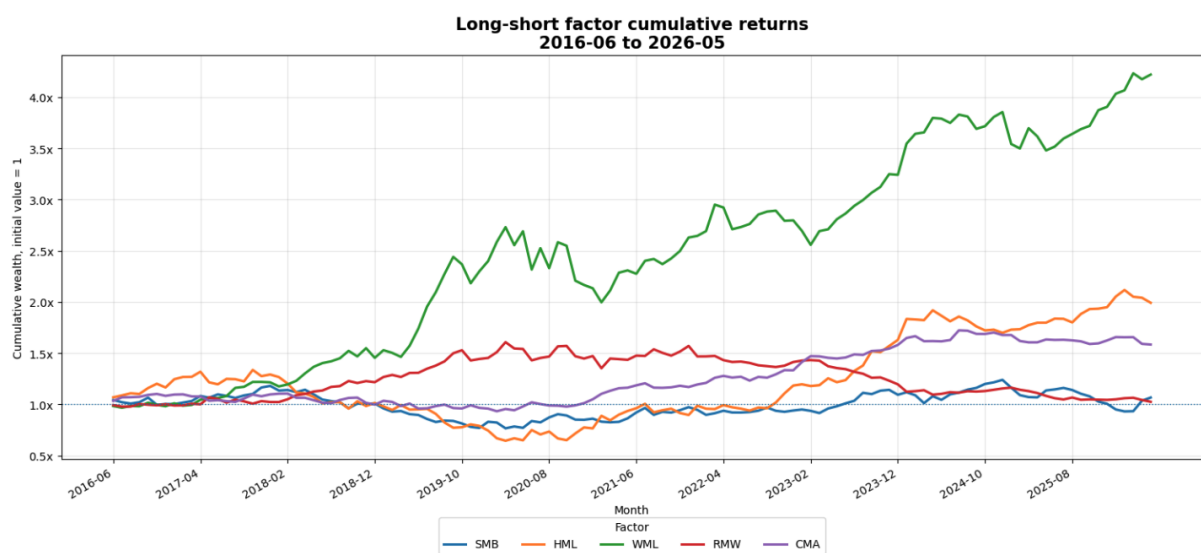


Figure 18: Each line represents long-short cumulative wealth starting from ₹1 using value weighted methodology

TABLE 1: Cross-sectional distribution of median annual stock price by portfolio year. Each cell reports the cross-sectional percentile of firm-level median daily prices within the portfolio year universe.

PY	P10	P25	P50	P90
2005	4.08	8.50	24.58	247.82
2010	4.90	12.50	36.95	323.70
2015	2.78	7.99	26.31	410.25
2020	1.70	6.30	22.75	349.38
2025	8.06	27.01	96.77	962.82

TABLE 2: Firm Counts by Portfolio Year: All Firms and Holding Firms

Portfolio Year	All Firms	Holding Firms
2001	3,539	858
2002	3,115	670
2003	2,926	705
2004	2,869	771
2005	3,010	1,097
2006	2,978	1,664
2007	3,107	1,772
2008	3,227	1,907
2009	3,241	2,079
2010	3,439	1,797
2011	3,540	2,162
2012	3,727	2,184
2013	3,782	1,995
2014	3,874	1,582
2015	4,047	1,692
2016	4,040	1,953
2017	4,139	1,971
2018	4,261	2,037
2019	4,327	2,118
2020	4,287	1,988
2021	4,295	1,862
2022	4,410	2,179
2023	4,574	2,622
2024	4,909	2,803
2025	5,231	3,126
2026	5,291	3,558

TABLE 3: Number of Firms Excluded Due to Negative Book Equity by Portfolio Year

PY	Negative BE Firms	PY	Negative BE Firms
2001	33	2013	81
2002	10	2014	60
2003	24	2015	90
2004	27	2016	128
2005	45	2017	122
2006	113	2018	116
2007	79	2019	129
2008	80	2020	99
2009	89	2021	71
2010	49	2022	98
2011	93	2023	134
2012	92	2024	117
		2025	146

Factor	Metric	Breakpoints	Labels
Op. Profitability	OpProf	30/70 pctile	W (Weak), N (Neutral), R (Robust)
Investment	Asset Growth	30/70 pctile	C (Conservative), N (Neutral), A (Aggressive)
Value	Book-to-Market	30/70 pctile	G (Growth), N (Neutral), V (Value)
Momentum	12-1 Mo Ret	30/70 pctile	L (Loser), N (Neutral), W (Winner)
Size	Market Cap	90th pctile	S (Small), B (Big)

TABLE 4: Factor Breakpoints and Labels

TABLE 5: Long and short legs for each factor return.

Factor	Long leg	Short leg	Label update
SMB	$SV + SN + SG$	$BV + BN + BG$	Annual (September)
HML	$SV + BV$	$SG + BG$	Annual (September)
WML	$SW + BW$	$SL + BL$	Monthly (both size and momentum)
RMW	$SR + BR$	$SW + BW$	Annual (September)
CMA	$SC + BC$	$SA + BA$	Annual (September)

TABLE 6: Factor Return Statistics: Full Period (2003:10–2024:09)

Factor	Statistic	Ours	JKP	AJV
SMB	Factor premium (% ann.)	4.17		0.86
	Monthly vol (%)	3.65		4.50
	Sharpe ratio	0.39		0.13
	Max drawdown (%)	−35.09		−56.04
HML	Factor premium (% ann.)	3.92	0.61	10.16
	Monthly vol (%)	5.18	5.37	5.19
	Sharpe ratio	0.30	0.12	0.63
	Max drawdown (%)	−68.86	−64.20	−50.32
WML	Factor premium (% ann.)	10.32	2.37	13.25
	Monthly vol (%)	6.11	6.59	5.60
	Sharpe ratio	0.58	0.24	0.74
	Max drawdown (%)	−60.28	−72.31	−51.96
RMW	Factor premium (% ann.)	4.01	1.15	
	Monthly vol (%)	3.08	4.47	
	Sharpe ratio	0.42	0.15	
	Max drawdown (%)	−34.52	−45.09	
CMA	Factor premium (% ann.)	5.10	−0.29	
	Monthly vol (%)	2.53	4.50	
	Sharpe ratio	0.61	0.07	
	Max drawdown (%)	−25.42	−56.02	

TABLE 7: 10-year long-only factor performance (June 2016 - May 2026)

Factor	Description	Cumulative	Annual Mean Returns	Annual Mean Excess	Volatility	Sharpe Ratio	Max Drawdown
Small	Size	4.2×	19.99%	14.37%	25.99%	0.55	-61.04%
Value	Value	5.4×	22.91%	17.30%	28.96%	0.60	-67.01%
Winner	Momentum	6.4×	22.45%	16.84%	21.19%	0.79	-29.95%
Robust	Profitability	1.9×	12.51%	6.90%	18.34%	0.38	-36.61%
Conservative	Operating Profitability	2.7×	15.54%	9.93%	21.63%	0.46	-55.34%
Nifty 500	Benchmark	2.3×	13.57%	7.96%	17.00%	0.47	-29.98%

TABLE 8: 10-year long-short factor performance (June 2016 - May 2026)

Factor	Description	Cumulative	Annual Mean	Annual Mean Excess	Volatility	Sharpe Ratio	Max Drawdown
SMB	Size	0.07×	1.42%	-4.18%	12.54%	-0.33	-35.09%
HML	Value	0.99×	8.32%	2.72%	17.02%	0.16	-51.76%
WML	Momentum	3.22×	15.70%	10.10%	15.63%	0.65	-26.89%
RMW	Operating Profitability	0.03×	0.70%	-4.90%	9.49%	-0.52	-36.22%
CMA	Investment	0.58×	4.91%	-0.69%	7.77%	-0.09	-15.60%